

DATE: 24 Dec. 2021, FN(8 -11) am

SECTION - A (ANALOG DESIGN)

M.M--35 TIME: 1 hr.

Attempt section -A, and Section -B in separate Answer sheets. Write the section name clearly on top of each sheet.

Answer the sub-parts of a question at one place and in the order in which they appear.

- Q1. A stable current source of 0.05mA is available in the design library. Using this current source, design a current source which can source 0.1mA over a wide range of voltage (80% of V_{DD}) and output resistance $> 20 \text{ MOhm}$. The supply voltage is 3V, threshold voltage of NMOS and PMOS are +0.8V and -0.8V respectively, $\mu_{cox}(\text{PMOS})=0.2 \text{ mA/V}^2$, $\mu_{cox}(\text{NMOS})=0.6 \text{ mA/V}^2$. Neglect body effect. Ignore channel length modulation while doing the DC analysis, for AC analysis use λ (NMOS and PMOS) = 0.01 /V. What is the power consumptions of the design? Draw the schematic, show the W/L of each transistor clearly. Use of resistors is not allowed.

[15 Marks]

- Q2. A Band Gap Reference (BGR) circuit (Fig. 1) is used to generate a reference voltage of 1.2V. Given the V_{EB} of the BJT is 0.6V, $R_1=R_2=10 \text{ k}\Omega$. Given V_T (thermal voltage) at room temperature is 26mV.

- a) Find the value of R_3 to generate 1.2V reference voltage ($V_{Ref.}$).
b) Assuming the resistor values (R_1 , R_2 and R_3), V_{EB} are independent of temperature, what should be the value of R_3 such that the variation in reference voltage ($V_{Ref.}$) is less than $40 \text{ mV}/^\circ\text{C}$.
c) What is the major drawback of this BGR circuit with respect to the range of $V_{Ref.}$.

[10 Marks]

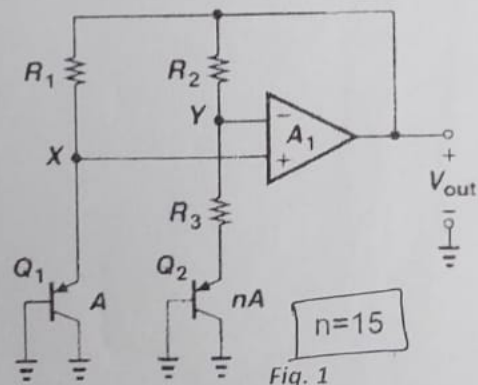


Fig. 1

- Q3. For the two stage op-amp shown in Fig. 2. Find the expression for low frequency differential voltage gain. Write expression for the magnitude of first three major poles.

[10 Marks]

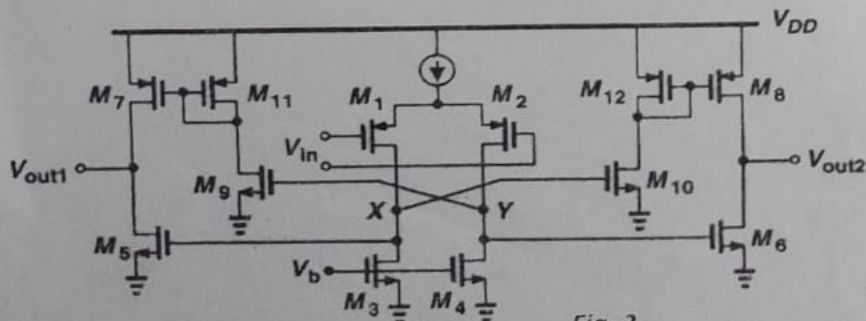


Fig. 2

-----End Section -A-----

General Instructions to the candidate

- Please write your answers legibly and neatly. Symbols, constants, terminology used have their usual meaning unless specified specially.
- There are a total of (four + three) questions and all are compulsory. Each carries marks as indicated

Common data: Use the following common data if not mentioned specifically in the question

For 0.25 μ m Technology node --

Take $V_{dd}=2.5$ V, $V_{TON}=0.4$ V, $V_{TOP} = -0.4$ V, $L_{min}=250$ nm, $\lambda_n=0.05$ V⁻¹, $\lambda_p=0.05$ V⁻¹, $\mu_n C_{ox}=120$ μ A/ V², $\mu_p C_{ox}=40$ μ A/ V²

- Use long channel approximation and lambda based design rules unless specially mentioned.
- Use square law current equation unless specifically mentioned. Neglect body effect, channel length modulation if not mentioned specifically. The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

SECTION - B DIGITAL DESIGN (M.M--70 , TIME: 2 hrs.)

Attempt section -A, and Section -B in separate Answer sheets. Write the section name clearly on top of each sheet.

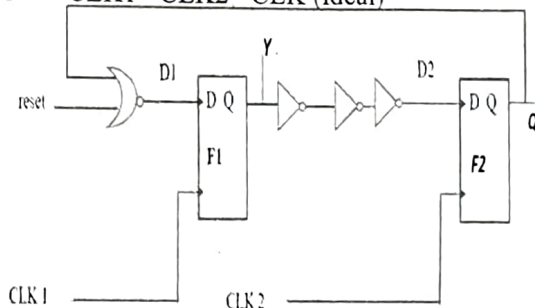
Answer the sub-parts of a question at one place and in the order in which they appear.

Q1. For the system shown in fig. 1, following data is given --- CLK1= CLK2= CLK (ideal)

D flop/register----- $t_{cq}=1$ ns, $t_{su}=2$ ns, $t_{hold}=2.5$ ns

NOR gate ----- $t_{cd}=2$ ns, $t_{pd-logic}=3$ ns

Inverter----- $t_{cd}=1$ ns, $t_{pd-logic}=2$ ns



a) Determine the maximum clock frequency for which circuit will operate correctly. Will the circuit function if two inverters are removed from the given circuit. ? Justify with reason.

b) Determine minimum clock time period and hold time of D-register for which circuit operates correctly if clk signal has a both positive as well as negative clock skew of value 1.5ns.

[15]

Q2. (a) Consider saturation enhancement-load NMOS inverter circuit of Fig. 2 . Given parameters are –

$|V_T|=0.8$ V, $\mu_n C_{ox}=102$ μ A/V², $(W/L)_{M3}=2$, $(W/L)_{M1}=20$, $V_{DD}=5$

V (Neglect body bias effect in calculations)

$(W/L)_{M2}=2$

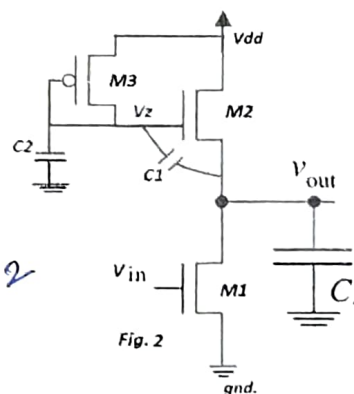


Fig. 2

a) Determine values of V_{OH} , V_{OL} , V_M . Hence sketch and label the voltage transfer characteristic (VTC)

b) Determine the ratio $(C2/ C1)$ to obtain full logic '1' output.

c) Determine the value of V_{OH} if transistor M2 is made PMOS. What is the drawback of this circuit? Explain.

[20]

Q3. Answer the following—

- A two terminal MOS structure was fabricated in 7 nm technology node using silicon dioxide (SiO_2) as an insulator. Further, it has been observed that this MOS structure was causing a huge gate leakage current just because of ultrathin gate oxide layer. In view of getting low gate leakage current SiO_2 layer was replaced by a 10 nm Hafnium dioxide (HfO_2) layer without compromising the key electrical parameters of the MOS capacitor. Find the thickness of SiO_2 and Maximum electric field in the new gate oxide. Consider Dielectric constant of $\text{SiO}_2=3.9$, Dielectric constant of $\text{HfO}_2=25$, and Maximum Electric field in $\text{SiO}_2=2.5 \times 10^4 \text{ V/cm}$
- Implement a 4:1 Multiplexer using minimum number of transistors. There is no constraint on the speed, power and voltage levels. The four input lines are D0,D1,D2 and D3 the select lines are S1,S0. Output is Y. All inputs are available only in true form.
- Modify and redraw the above 4:1 MUX this time in addition to the transistor count, it is necessary to reduce the power and have full voltage swing at the output node. Mention a note on the constraints of this additional transistors added.

[15]

Q4. Consider the circuit schematics as shown in Fig.4(a) & Fig.4(b). The applied input signal/s V_{IN} has frequency of **100MHz** and the rise/fall time of the input signal is **10 pico-sec**. For transistors M3, and M4, threshold voltage is $|V_t|=0.6\text{V}$. Assume full logic levels of input signal. Neglect parasitic capacitances. Assume same load capacitance at output

- Now, analyze both the circuits and determine their function/operation.
- Size the width of transistors to have worst case (symmetrical) response using method of logical effort.. Given $p_{inv}=1$, $g_{inv}=1$ for reference cmos inverter.

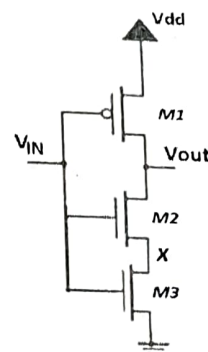


Fig. 4a

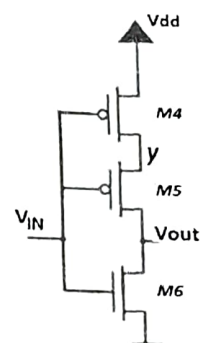


Fig. 4b

- Determine, which circuit has lower dynamic power dissipation per clock cycle. Justify with proper reason.
- Compare, qualitatively, both the circuits and find out which circuit is better in terms propagation delay. Justify your answer.
- Now, calculate the short circuit power dissipation per clock cycle of each circuit. Now, suppose the pull up block of Fig.4a is replaced by pull up block of Fig. 5b. . Does it reduce short circuit power consumption? Justify your answer with proper reason/s.

[20]

-----End Section -B-----